

# Loan Approval Prediction

## Alfido Tech Data Science Internship – Task 2

### 1. Project Overview

This project was completed as part of the Alfido Tech Data Science Internship.

The objective of this project is to perform Exploratory Data Analysis (EDA) on a Loan Approval dataset and understand patterns related to loan approval status, CIBIL scores, annual income, and loan amounts.

### 2. Dataset Overview

Dataset: Loan Approval Prediction Dataset

Source: Kaggle

Total Applicants: 4,269

Number of Columns: 13

Target Column: loan\_status

The dataset contains information about applicants such as dependents, education, employment status, annual income, loan amount, loan term, CIBIL score, assets, and loan status.

### 3. Tools and Technologies

Python

Pandas

Matplotlib

Google Colab

Exploratory Data Analysis (EDA)

#### 4. Analysis Performed

The following analysis was performed:

Loaded and inspected the dataset.

Checked the dataset shape and column names.

Checked data types and missing values.

Generated descriptive statistics.

Analyzed loan approval status distribution.

Analyzed the distribution of CIBIL scores.

Examined the relationship between annual income and loan amount.

Compared CIBIL scores across loan approval statuses.

Generated four visualizations to understand the dataset.

#### 5. Visualizations

##### 6. Loan Approval Status Distribution

A bar chart showing the distribution of approved and rejected loan applications.

##### 2. CIBIL Score Distribution

A histogram showing the distribution of CIBIL scores among applicants.

##### 3. Annual Income vs Loan Amount

A scatter plot showing the relationship between applicants' annual income and requested loan amount.

## 5. CIBIL Score by Loan Status

A box plot comparing CIBIL score distributions across loan approval statuses.

Important: Nee Colab graphs ni screenshots ga teesukoni ikkada respective sections kinda paste cheyyi.

## 6. Key Insights

Total applicants: 4,269

Approved applications: 2,656

Most common loan status: Approved

Average CIBIL score: 599.94

Average annual income: ₹5,059,123.92

Average loan amount: ₹15,133,450.46

CIBIL score range: 300–900

## 7. Conclusion

The analysis provides an overview of loan applicants and their approval outcomes. The dataset contains 4,269 applications, with Approved being the most common loan status.

CIBIL score, annual income, and loan amount were explored using descriptive statistics and visualizations. The analysis helps understand the characteristics of loan applicants and provides a foundation for further machine-learning-based loan approval prediction.

## 8. Project Deliverable

The complete Python notebook contains the data loading, data inspection, EDA, visualizations, and key insights.